



FairJudge: An Adaptive, Debiased, and Consistent LLM-as-a-Judge

Bo Yang^{*1} Lanfei Feng^{*2} Yunkui Chen² Xiao Xu¹ Yu Zhang¹ Shijian Li¹

Abstract

Existing LLM-as-a-Judge systems suffer from three fundamental limitations: **limited adaptivity** to task and domain-specific evaluation criteria, **systematic biases** driven by non-semantic cues such as position, length, format, and model provenance, and **evaluation inconsistency** that leads to contradictory judgments across different evaluation modes (e.g., pointwise versus pairwise). To address these issues, we propose **FairJudge**, an adaptive, debiased, and consistent LLM-as-a-Judge. Unlike prior approaches that treat the judge as a static evaluator, FairJudge models **judging behavior itself as a learnable and regularized policy**. From a data-centric perspective, we construct a high-information-density judging dataset that explicitly injects supervision signals aligned with evaluation behavior. Building on this dataset, we adopt a curriculum-style SFT-DPO-GRPO training paradigm that progressively aligns rubric adherence, bias mitigation, and cross-mode consistency, while avoiding catastrophic forgetting. Experimental results on multiple internal and public benchmarks show that FairJudge improves agreement and F1 across several evaluation settings, reduces selected non-semantic biases, and achieves competitive or stronger performance than larger general-purpose LLMs on judge-oriented tasks.

1. Introduction

Large language models are increasingly used as automatic judges in model evaluation, preference learning, and supervision pipelines, and have gradually become a critical

^{*}Equal contribution ¹College of Computer Science, Zhejiang University, Hangzhou, China ²School of Software Technology, Zhejiang University, Hangzhou, China. Correspondence to: Shijian Li <shijianli@zju.edu.cn>.

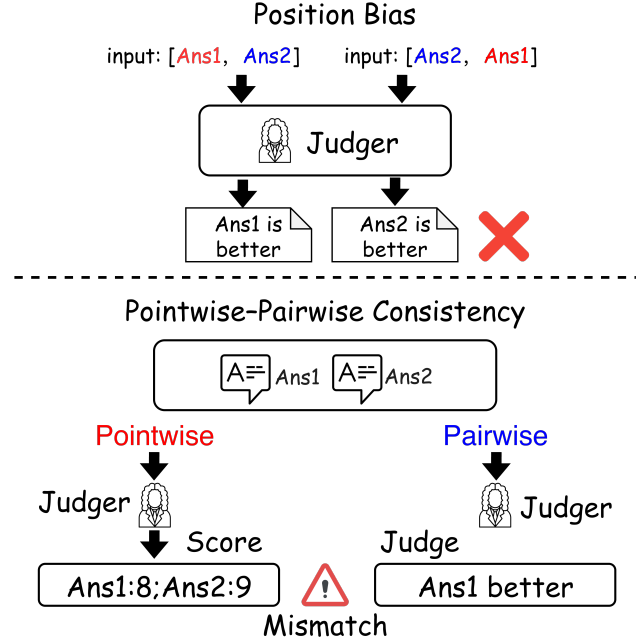


Figure 1. Motivation—Two representative issues in LLM-as-a-Judge. Top: **Position bias**, where the judgment flips when the order of answers is swapped. Bottom: **Pointwise-pairwise inconsistency**, where the same answers receive contradictory judgments under different evaluation modes. Representative real-world examples are provided in Appendix Figure 6.

infrastructure component of modern machine learning systems (Zheng et al., 2023; Chiang et al., 2024). Recent studies have proposed dedicated judge models, datasets, and evaluation benchmarks, demonstrating that targeted training can substantially improve automatic evaluation under specific settings (Zhu et al., 2023; Wang et al., 2023; Christie et al., 2024; Liu et al., 2023; Kocmi & Federmann, 2023). However, as illustrated in Figure 1, in more complex real-world evaluation scenarios, existing LLM-based judges still exhibit notable instability and limited generalization, which constrains their reliability as general-purpose evaluators (Tripathi et al., 2025; Shi et al., 2025; Wang et al., 2024).

These limitations are not merely a consequence of insufficient model capacity or data scale, but reflect a deeper

challenge: judging is not a static prediction task, but a behavior decision process jointly constrained by evaluation rules, context, and evaluation settings (Amodi et al., 2016; Hadfield-Menell et al., 2017; Christiano et al., 2017). In practice, evaluation criteria vary across tasks and domains, judgments should be robust to non-semantic perturbations such as answer order, length, or formatting, and evaluation outcomes should remain self-consistent across different evaluation modes (e.g., pointwise versus pairwise) (Tripathi et al., 2025; Shi et al., 2025; Wang et al., 2024). Nevertheless, existing approaches often implicitly assume these properties to emerge naturally, rather than treating them as explicit learning objectives (Ouyang et al., 2022; Bai et al., 2022; Rafailov et al., 2023; Gao et al., 2023).

We argue that the root cause of these issues lies in the fact that judging behavior itself has not been systematically modeled as a learnable and optimizable objective. While prior work has introduced task-specific judge models and datasets, adaptivity, fairness, and consistency in judging are typically assumed rather than jointly constrained during learning (Zhu et al., 2023; Wang et al., 2023; Liu et al., 2023). As a result, LLM-based judges often fail to maintain stable and reliable judgments under rule changes, non-semantic perturbations, or evaluation setting shifts (Tripathi et al., 2025; Wang et al., 2024; Rafailov et al., 2023).

Based on this observation, we propose **FairJudge**, a method that explicitly treats judging behavior as a learning objective, aiming to systematically improve LLM-as-a-Judge systems in terms of adaptivity, debiasing, and consistency. (Gehrmann et al., 2021; Celikyilmaz et al., 2020; Bowman, 2024; Lin, 2004; Papineni et al., 2002; Achiam et al., 2023; Gilardi et al., 2023).

Our main contributions are summarized as follows:

- **FairJudge.** We propose a new perspective that treats judging behavior as a learnable and optimizable objective, and introduce **FairJudge** accordingly, together with the publicly released **FairJudge-16K** training dataset and **FairJudge-Benchmark-1K** evaluation benchmark.
- **Adaptive Judging.** FairJudge explicitly improves the adaptivity of LLM-based judges to task- and domain-specific evaluation criteria by enabling robust alignment with changing rules and contextual requirements.
- **Debiased Judging.** FairJudge systematically mitigates non-semantic biases induced by factors such as answer order, length, and formatting, leading to fairer and more reliable evaluation outcomes.
- **Consistent Judging.** FairJudge significantly enhances judgment consistency across different evaluation settings, including pointwise and pairwise modes, im-

proving the practical reliability of LLM-as-a-Judge systems.

Conflict of Interest Disclosure. The authors declare no financial conflicts of interest or other substantive conflicts that could reasonably be perceived to influence the work.

2. Related Work

2.1. LLM-as-a-Judge and Automated Evaluation

LLM-as-a-Judge has emerged as a practical paradigm for model evaluation and preference-based alignment. Benchmark-driven frameworks such as MT-Bench and Chatbot Arena establish scalable evaluation protocols based on human preferences and pairwise comparisons (Zheng et al., 2023; Chiang et al., 2024). Beyond benchmarks, several works train or adapt LLMs as dedicated judges under fixed evaluation settings, including JudgeLM (Zhu et al., 2023), PandaLM (Wang et al., 2023), and tool-based evaluation systems such as FlexEval (Christie et al., 2024). More recent approaches explore fine-grained or generative judging with explicit rubric conditioning, exemplified by Prometheus (Kim et al., 2023), Generative Judge (Auto-J) (Li et al., 2023), and multilingual judge suites such as M-Prometheus (Pombal et al., 2025). Surveys provide an overview of this rapidly growing area and summarize open challenges in robustness and generalization (Li et al., 2024; Gu et al., 2024).

2.2. Adaptive, Debiased, and Consistent Judging

A growing body of work has revealed that LLM-based judges exhibit systematic limitations in adaptivity, debiasing, and consistency. Judgments may be influenced by non-semantic factors such as answer position, length, and formatting, resulting in biased or unfair evaluation outcomes (Wang et al., 2024; Shi et al., 2025; Wang et al., 2025; Zhang et al., 2025; Li et al., 2025; Liu et al., 2024b; Anghel et al., 2025). Length bias, in particular, has been shown to significantly affect automatic preference evaluation, motivating explicit debiasing strategies such as Length-Controlled AlpacaEval (Dubois et al., 2024), as well as additional mitigation analyses (Zhou et al., 2024). Beyond bias, recent studies demonstrate that different evaluation protocols can yield contradictory judgments: pointwise and pairwise evaluation modes may diverge under identical content due to protocol-specific sensitivities (Tripathi et al., 2025). These findings indicate that existing LLM judges often lack robust adaptation to rule changes, effective debiasing against non-semantic cues, and consistent behavior across evaluation settings (Li et al., 2024; Gu et al., 2024).

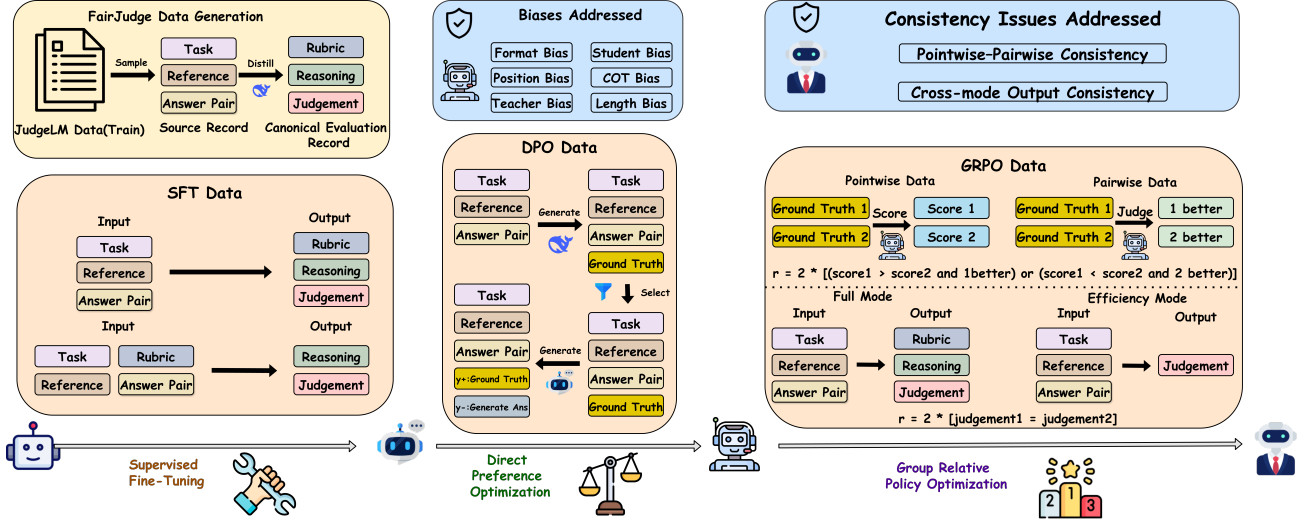


Figure 2. Data construction pipeline of FairJudge. **Left (SFT data):** Source evaluation records are organized into {Task, Reference, Answer Pair} and augmented with explicit {Rubric, Reasoning, Judgement}. The rubric is used both as a conditioning input and, in selected cases, as a prediction target. **Middle (DPO data):** Preference pairs (chosen/rejected) are constructed on the same evaluation instance under targeted non-semantic perturbations, guiding the judge to be robust to non-semantic biases. **Right (GRPO data):** Cross-mode samples are organized by jointly constructing **pointwise** (scores or labels) and **pairwise** (relative preference) evaluations for the same instance, with consistency rewards aligning judgments across modes to enforce **cross-mode consistency**.

2.3. Preference Learning and Training Judges

From a training perspective, LLM judges are closely related to preference learning and alignment methods. RLHF establishes a foundational paradigm for aligning models with human preferences (Ouyang et al., 2022; Bai et al., 2022; Christiano et al., 2017), while Direct Preference Optimization (DPO) provides a simplified alternative without explicit reward model training (Rafailov et al., 2023). Large-scale AI feedback datasets such as UltraFeedback support scalable preference supervision (Cui et al., 2023), and benchmarks such as RewardBench enable systematic evaluation of reward and judge models (Lambert et al., 2025). At the same time, prior work highlights risks of reward overoptimization and generalization failures, underscoring the need for principled constraints when training judges (Gao et al., 2023).

3. Methodology

3.1. Problem Formulation: Judging as a Learnable Decision Policy

In most existing LLM-as-a-Judge approaches, the judging process is implicitly modeled as a deterministic mapping from evaluation inputs to output judgments. In the common pointwise setting, a judge directly assigns a score or label to a given input:

$$\text{Judge}_{\text{pt}}(x) \rightarrow y,$$

where x denotes the evaluation input (e.g., a question-answer pair) and y represents the final judgment, such as a score, label, or preference. In pairwise evaluation, judging is typically formulated as a comparison function over two candidate answers:

$$\text{Judge}_{\text{pw}}(x_1, x_2) \rightarrow y,$$

where y indicates the relative preference between the two inputs.

Under this function-based formulation, evaluation rules and decision logic are implicitly encoded in model parameters or prompts. Differences across tasks, rubrics, or evaluation protocols (e.g., pointwise versus pairwise) are usually handled through mode-specific prompts. As a result, the judging behavior is largely fixed and difficult to control systematically across varying evaluation settings.

In practice, judging behavior depends not only on the input content itself, but also critically on the evaluation context, including task-specific criteria, scoring rubrics, and evaluation modes. The same answer may warrant different yet logically consistent judgments under different rules or protocols. Modeling judging as a single fixed function makes it difficult to capture this conditional dependence and often leads to instability, bias, and inconsistency across settings.

To address this limitation, we explicitly model judging as a *conditional decision policy*:

$$\pi_{\text{judge}}(y \mid x, c, m),$$

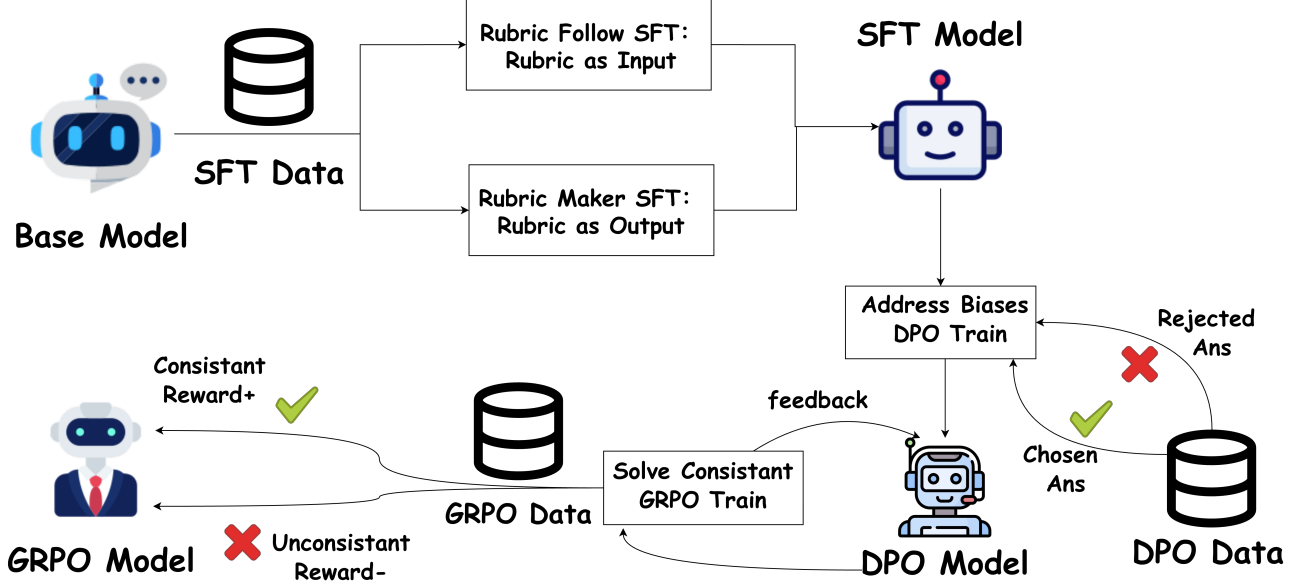


Figure 3. Training pipeline of FairJudge. The model is first trained with **SFT data**, where rubrics are used as conditioning inputs and, in some cases, as prediction targets, enabling explicit modeling of evaluation criteria. It is then optimized with **DPO data** in the form of chosen/rejected judgment pairs to reduce non-semantic biases. Finally, **GRPO training** applies consistency-oriented rewards, encouraging judgments that remain stable across evaluation settings. This staged process yields a rubric-aware, debaised, and consistent judge.

where x denotes the evaluation input, c represents the evaluation context (e.g., rubric or reference), m denotes the evaluation mode (such as pointwise or pairwise), and y is the judgment output. Under this formulation, the output y is viewed as a sample from the judging policy conditioned on the evaluation setting, rather than as the unique output of a fixed scoring function.

This policy-based perspective shifts the modeling focus from individual judgment outputs to the overall *behavior* of the judge across conditions. Specifically, the goal of the judging system becomes learning a policy that behaves in a stable, controllable, and generalizable manner under varying contexts and evaluation modes.

In FairJudge, we further characterize the quality of a judging policy along three key dimensions: (1) **Adaptivity**, the ability to adjust judgments in response to different evaluation criteria; (2) **Debiasing**, robustness against non-semantic perturbations such as answer order, length, or formatting; and (3) **Consistency**, the preservation of coherent judgment logic across different evaluation modes. As illustrated in Figure 4, this policy-based formulation provides a principled foundation for subsequent data construction and training, enabling explicit constraints on judging behavior rather than implicit reliance on prompt design. Representative real execution examples of our model are provided in Figure 7.

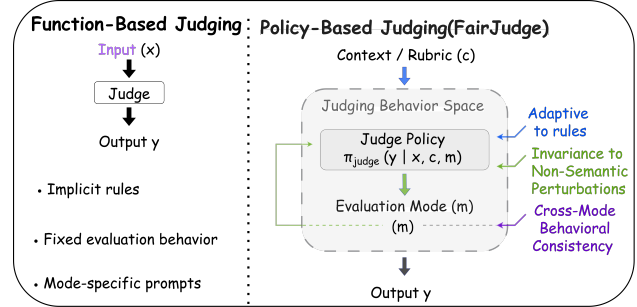


Figure 4. Comparison between function-based judging and policy-based judging.

3.2. Data Construction for Policy-Oriented Judging

To support modeling judging behavior as a learnable and controllable policy, FairJudge does not simply collect preference pairs or scalar scores. Instead, we construct a high-information-density training dataset. Representative examples of the various training data used are provided in Appendix Figure 5.

Canonicalizing Judging Records. As shown in Figure 2, FairJudge-16K is constructed from JudgeLM-style evaluation records (Zhu et al., 2023), rather than directly from raw model outputs. Detailed statistics of data distribution and filtering procedures corresponding to Figure 9, Figure 8, 10, and 11 are provided in the Appendix.

Table 1. Comparison of LLM-as-a-Judge performance across three evaluation benchmarks. The table compares general-purpose multimodal models and specialized judge models on three test sets. PandaLM is a human-annotated benchmark, reflecting alignment with human judgment preferences, while JudgeLM and FairJudge-Benchmark-1K evaluate model robustness and consistency in automated judging scenarios. Baselines include both general LLMs and dedicated LLM-as-a-Judge methods (marked with *). We further report FairJudge results at three model scales (2B / 4B / 8B), all initialized from the corresponding Qwen3-VL backbones, to analyze how judging performance scales with model capacity. The best result in each column is highlighted in **bold**.

Model	Agreement	Precision	Recall	F1
PandaLM Test Set (Human Annotations)				
Qwen3-8B	70.05	63.69	64.80	63.79
GLM-4-9B	70.02	<u>69.86</u>	57.69	59.55
InternVL3-14B	66.67	<u>65.07</u>	68.67	62.27
LLaVA-1.5-7B	37.13	40.11	39.24	28.98
LLaVA-1.5-13B	29.89	43.16	41.67	28.28
Qwen2.5-72B	<u>72.80</u>	66.25	63.60	64.58
DeepSeek-V3-671B	64.76	58.48	61.13	58.83
FlexVL-7B*	71.34	64.30	62.18	63.00
JudgeLM-7B*	66.77	63.83	<u>71.95</u>	63.92
PandaLM-7B*	65.82	44.56	48.93	46.24
FairJudge-2B	71.53	67.75	59.99	61.68
FairJudge-4B	69.31	64.65	61.02	62.18
FairJudge-8B	76.83	71.87	72.54	72.18
JudgeLM Test Set				
Qwen3-8B	72.15	58.77	58.78	58.72
GLM-4-9B	72.42	59.06	53.31	52.54
LLaVA-1.5-7B	40.80	43.11	41.16	34.34
LLaVA-1.5-13B	48.59	41.61	40.45	39.69
InternVL3-14B	71.96	62.26	64.69	61.23
Qwen2.5-72B	79.59	71.31	60.95	61.92
DeepSeek-V3-671B	72.04	59.38	60.37	59.75
FlexVL-7B*	<u>77.44</u>	64.82	57.49	57.42
JudgeLM-7B*	78.00	<u>66.02</u>	67.61	<u>66.66</u>
PandaLM-7B*	67.42	44.99	48.53	46.66
FairJudge-2B	71.20	59.88	52.36	51.48
FairJudge-4B	75.07	71.03	59.59	60.95
FairJudge-8B	<u>78.82</u>	66.77	<u>66.93</u>	66.78
FairJudge-Benchmark-1K				
Qwen3-8B	63.93	62.85	56.45	58.35
GLM-4-9B	64.40	60.79	56.29	57.75
InternVL3-14B	58.31	57.76	62.30	55.11
LLaVA-1.5-7B	40.34	37.42	35.49	32.57
LLaVA-1.5-13B	45.86	38.93	41.95	39.03
Qwen2.5-72B	<u>67.45</u>	<u>69.23</u>	57.39	59.72
DeepSeek-V3-671B	55.97	49.97	51.66	50.26
FlexVL-7B*	59.58	49.56	46.01	44.72
JudgeLM-7B*	69.56	65.27	67.41	<u>66.11</u>
PandaLM-7B*	52.46	35.00	39.23	36.99
FairJudge-2B	62.17	58.89	58.15	56.37
FairJudge-4B	67.74	72.19	59.97	62.57
FairJudge-8B	71.50	71.15	<u>66.92</u>	67.63

Table 2. Ablation Study of FairJudge. We evaluate the effect of removing different training stages (SFT, DPO, GRPO) on three test sets.

Model	Agreement	Precision	Recall	F1
PandaLM Test Set (Human Annotations)				
BASE(Qwen3-VL-8B)	69.63	54.31	59.84	54.05
W/O SFT	74.07	67.85	70.42	68.74
W/O DPO	75.70	70.59	69.91	70.05
W/O GRPO	74.50	68.53	71.21	69.45
FairJudge-8B	76.83	71.87	72.54	72.18
JudgeLM Test Set				
BASE(Qwen3-VL-8B)	73.84	59.89	60.51	59.93
W/O SFT	76.32	63.80	64.14	63.86
W/O DPO	77.58	65.56	67.17	66.11
W/O GRPO	76.78	64.13	64.02	63.95
FairJudge-8B	78.82	66.77	66.93	66.78
FairJudge-Benchmark-1K				
BASE(Qwen3-VL-8B)	66.35	59.65	59.54	59.54
W/O SFT	68.62	71.00	56.12	57.86
W/O DPO	68.15	64.12	58.13	59.64
W/O GRPO	68.85	72.32	57.39	59.69
FairJudge-8B	71.50	71.15	66.92	67.63

Each source record is restructured into a canonical evaluation instance consisting of a *task*, *reference information*, *answer pair*, *rubric*, *reasoning*, and *final judgment*. This canonicalization explicitly exposes the contextual conditions under which judgments are made, transforming previously implicit evaluation criteria into learnable signals.

Structured Construction for Debiasing. To address non-semantic biases commonly observed in LLM-based judges, FairJudge-16K introduces systematic contrastive constructions at the data level (middle of Figure 2). While preserving semantic equivalence, evaluation inputs are perturbed along non-semantic dimensions such as answer order, length, formatting, reasoning style, and teacher model provenance. These structured perturbations explicitly constrain the judging policy to be invariant to non-semantic factors, allowing debiasing to be learned as a behavioral property rather than enforced through prompt heuristics.

Cross-Mode Consistency Supervision. Beyond bias mitigation, FairJudge-16K explicitly targets consistency across evaluation modes. As illustrated on the right side of Figure 2, the same evaluation content is paired with both pointwise and pairwise supervision, enabling direct alignment between different judgment formats. This cross-mode pairing provides the necessary foundation for consistency-oriented optimization in later training stages, preventing contradictory judgments across evaluation protocols.

3.3. FairJudge-Benchmark-1K

During the construction of the FairJudge training data, we simultaneously curated and froze a scale-controlled evaluation subset, referred to as **FairJudge-Benchmark-1K**. Rather than being created post hoc after model training, this benchmark naturally emerged as a by-product of the data generation pipeline, and is designed to characterize judge behavior under realistic evaluation conditions.

Specifically, FairJudge-Benchmark-1K is independently sampled from the normalized evaluation records, covering diverse task types, evaluation rubrics, and evaluation modes (e.g., pointwise and pairwise). Each instance preserves the full evaluation context, candidate answer pairs, and their canonicalized judgments, enabling systematic analysis of adaptivity, debiasing, and cross-mode consistency in LLM-based judges.

To ensure evaluation reliability and strict separation from training, we incorporate a **human-in-the-loop** auditing process during the construction and freezing of this subset. Human inspection is limited to verifying structural integrity, consistency of evaluation metadata, and potential data leakage risks, without re-annotating or subjectively modifying judgment outcomes. Once verified, the benchmark is fully frozen and excluded from all stages of model training, preference optimization, and policy learning.

Since FairJudge-Benchmark-1K shares the same unified data generation and normalization pipeline as the training corpus, it remains distributionally aligned in terms of task structure and evaluation format, while being strictly disjoint at the instance level. This design allows the benchmark to reliably reflect judge performance in practical evaluation scenarios without introducing additional assumptions or confounding factors.

3.4. Training Paradigm: Curriculum Optimization of Judging Policies

As illustrated in Figure 3, FairJudge is trained through a three-stage curriculum that progressively aligns the judging policy with explicit evaluation rules, debiasing constraints, and cross-mode consistency requirements. Rather than treating training as a generic SFT–DPO–RL pipeline, each stage in FairJudge introduces a distinct and necessary supervision signal that targets a specific property of judging behavior.

Stage I: Supervised Fine-Tuning for Rubric-Aware Judging. The first stage aims to establish a stable and controllable judging behavior space. We perform supervised fine-tuning (SFT) using high-quality canonical evaluation records. Each training instance includes the evaluation task, reference context, candidate answer(s), and an explicit rubric. The model is trained to generate structured

judgments that explicitly condition on the provided rubric and context.

This stage serves two purposes. First, it enforces rubric-following behavior by making evaluation criteria an explicit part of the model input. Second, it stabilizes the output format (e.g., reasoning followed by judgment), which is essential for subsequent preference-based optimization. At this stage, the judge learns *how to judge*, but not yet *how to judge robustly*.

Stage II: Debiasing via Direct Preference Optimization.

While SFT aligns the judge with evaluation rules, it does not prevent the model from exploiting non-semantic cues such as answer length, position, or formatting. To mitigate such biases, we introduce a debiasing stage based on Direct Preference Optimization (DPO). Details about DPO are provided in Appendix A.1.

We construct preference pairs where the *chosen* judgment adheres to the rubric while remaining invariant to non-semantic variations, and the *rejected* judgment exhibits systematic bias. DPO then optimizes the judging policy to prefer unbiased evaluation behaviors without requiring an explicit reward model. This stage reduces sensitivity to spurious cues while preserving rubric adherence learned during SFT.

Stage III: Consistency Optimization via Group-Relative Policy Optimization.

The final stage addresses a critical but underexplored challenge: maintaining consistent judgments across different evaluation modes, such as pointwise and pairwise settings. Instead of relying on prompt engineering or post-hoc calibration, FairJudge explicitly encodes cross-mode consistency as an optimization objective.

We adopt Group-Relative Policy Optimization (GRPO), where multiple judgments are sampled for the same evaluation instance under different modes. A consistency-aware reward assigns higher scores to judgment groups that yield logically equivalent outcomes across modes, and penalizes contradictory decisions. By optimizing group-relative advantages, GRPO encourages stable and self-consistent judging behavior while avoiding reward overfitting. Details about GRPO objective and GRPO reward are provided in Appendix A.2 and Appendix A.3.

Overall Objective. The full training objective combines the three stages in a curriculum manner:

$$\mathcal{L} = \mathcal{L}_{\text{SFT}} + \lambda_{\text{DPO}} \mathcal{L}_{\text{DPO}} + \lambda_{\text{GRPO}} \mathcal{L}_{\text{GRPO}}, \quad (1)$$

where each component corresponds to a distinct behavioral constraint. Together, this curriculum enables FairJudge to learn a judging policy that is adaptive to evaluation rules, robust to non-semantic biases, and consistent across evaluation protocols.

Table 3. **Multimodal Benchmarks (MLLM-as-a-Judge, Human Annotations).** Accuracy (%) on diverse multimodal evaluation suites.

Model	COCO	Cha.QA	Co.Cap.	Visit.	WIT	Di.DB	Infog.	LLaVA.	MathV.	T.vqa	Avg
Qwen3-VL-8B	73.17	47.62	73.03	77.46	67.86	53.06	76.92	73.61	65.07	64.52	67.23
InternVL3-8B	67.07	38.10	69.66	62.86	58.93	53.06	61.04	70.83	45.89	51.61	57.91
LLaVA-1.5-7B	24.53	25.00	17.19	22.45	27.91	12.50	41.94	20.93	27.27	39.58	25.93
InternVL3-14B	50.00	39.68	62.92	61.43	48.21	67.35	74.03	62.50	50.00	50.00	56.61
Qwen3-VL-30B	69.51	36.51	67.42	73.24	64.91	83.67	71.43	70.83	58.22	69.35	66.51
FlexVL-7B*	75.61	57.14	74.39	74.39	72.73	47.92	84.13	67.14	59.85	70.97	68.49
JudgeLM*	36.59	22.22	53.93	45.07	54.39	24.49	9.09	43.06	23.97	17.74	33.05
PandaLM*	63.41	52.38	60.67	66.20	61.40	63.27	62.34	68.06	50.68	56.45	60.49
FairJudge-8B	78.05	53.97	69.66	74.29	76.79	61.22	79.22	70.83	65.07	66.13	69.52

Table 4. **Pointwise-Pairwise Consistency Scores on FairJudge-Benchmark-1K.**

Model	Consistency (%)
InternVL3-8B	48.52
GLM-4-9B	51.71
Qwen3-8B	54.21
Qwen3-VL-8B	60.59
InternVL3-14B	53.08
Qwen2.5-72B	53.76
FlexVL-7B*	46.00
DeepSeek-V3-671B	52.85
FairJudge-8B	65.52

4. Experiments and Results

Benchmarks. We evaluate LLM-as-a-Judge models on three primary test sets: the **PandaLM Test Set (Human Annotations)** (Wang et al., 2023), the **JudgeLM Test Set** (Zhu et al., 2023), and **FairJudge-Benchmark-1K**. In addition, multimodal judging performance is evaluated on the **MLLM-as-a-Judge benchmark** (Chen et al., 2024a), which aggregates multiple vision-language evaluation tasks into a unified test suite with human annotations.

Baselines. We compare FairJudge with two groups of baselines. *General-purpose models* include InternVL (Chen et al., 2024b), Qwen2.5-VL (Bai et al., 2025), LLaVA (Liu et al., 2024a), as well as larger general-purpose models where available, including **Qwen2.5-72B**, **Qwen3-VL-30B**, and **DeepSeek-V3-671B**. *Judge-oriented models* include PandaLM (Wang et al., 2023), JudgeLM (Zhu et al., 2023), and Flex-Judge (Ko et al., 2026). FairJudge is evaluated at three scales (2B/4B/8B), each trained from the corresponding Qwen3-VL backbone.

Evaluation Metrics. We report **Agreement**, **Precision**, **Recall**, and **F1** scores for all experiments. Agreement measures the overall consistency between model predictions

Table 5. **Reward-Bench Results.** Performance comparison on Reward-Bench using Agreement, Precision, Recall, and F1.

Model	Agreement	Precision	Recall	F1
LLaVA-1.5-7B	42.07	39.57	30.29	25.73
Qwen3-8B	77.25	53.17	51.67	52.11
Qwen3-VL-8B	82.08	55.99	54.71	55.34
InternVL3-8B	76.38	54.91	50.96	52.80
GLM-4-9B	68.24	46.68	45.54	46.05
LLaVA-1.5-13B	40.28	36.12	25.71	28.36
InternVL3-14B	75.23	58.26	50.03	53.70
DeepSeek-V3-671B	83.34	57.29	55.48	56.33
FlexVL-7B*	75.55	50.49	50.41	50.41
JudgeLM-7B*	46.57	41.43	30.89	35.27
PandaLM-7B*	53.00	54.53	53.77	51.30
FairJudge-8B	84.79	58.14	56.34	56.94

and reference annotations. The F1 score is computed as a macro-F1 over three judgment categories (*A*, *B*, *tie*), by averaging per-class F1 scores rather than deriving it from aggregated Precision and Recall.

We analyze the performance of FairJudge from five complementary perspectives: overall judgment quality, contributions of different training stages, cross-mode consistency, reward modeling capability, and inference efficiency.

4.1. Main Results: Structured Training Improves Judging Reliability

As shown in Table 1, FairJudge achieves consistently strong performance across all three evaluation benchmarks. In particular, on the PandaLM test set with human annotations, FairJudge substantially outperforms comparably sized general-purpose multimodal models in both Agreement and macro-F1, indicating better alignment with human judgment preferences.

Importantly, these gains cannot be attributed solely to model scale. Larger general-purpose models such as Qwen2.5-72B

Table 6. Inference Efficiency Wall-clock time measured on a single RTX 4090 GPU with 1,000 evaluation samples. The *Full* mode outputs complete reasoning and judgment, while the *Fast* mode outputs only the final decision.

Mode	Time	Speedup
PandaLM(Full Mode)	18 min 40 s	–
JudgeLM(Full Mode)	16 min 46 s	–
FairJudge-8B (Full Mode)	16 min 52 s	–
FairJudge-8B (Fast Mode)	1 min 19 s	$\times 12.8$
FairJudge-4B (Full Mode)	8 min 24 s	–
FairJudge-4B (Fast Mode)	39 s	$\times 12.9$
FairJudge-2B (Full Mode)	4 min 5 s	–
FairJudge-2B (Fast Mode)	18 s	$\times 13.6$

and DeepSeek-V3-671B do not exhibit uniformly superior judging performance, and their results vary considerably across datasets. In contrast, FairJudge achieves strong performance across the 2B, 4B, and 8B settings, suggesting that judge-specific data construction and staged training provide benefits beyond parameter scale alone. We do not interpret these results as a strictly monotonic scaling trend, since different benchmarks and metrics exhibit heterogeneous behavior across model sizes.

Compared with prior judge-oriented models (e.g., PandaLM, JudgeLM, and FlexVL), FairJudge achieves higher or comparable scores across the evaluated benchmarks, suggesting that judge-specific training generalizes across these settings rather than only fitting a single evaluation setup.

4.2. Ablation Study: Complementary Effects of Staged Training

The ablation results in Table 2 show that the three training stages contribute complementary improvements to FairJudge. Across all three benchmarks, the full FairJudge-8B model achieves the best Agreement and macro-F1, indicating that no single reduced variant fully matches the complete SFT–DPO–GRPO training pipeline.

A more careful reading of the results suggests that SFT plays the most fundamental role in establishing overall judging ability. Removing SFT leads to the largest macro-F1 degradation on all three benchmarks: from 72.18 to 68.74 on PandaLM, from 66.78 to 63.86 on JudgeLM, and from 67.63 to 57.86 on FairJudge-Benchmark-1K. This indicates that rubric-aware supervised training is essential for learning the basic judgment format, task conditioning, and alignment with reference annotations.

DPO and GRPO further improve the model beyond this supervised foundation, but their effects are reflected differently across metrics and datasets. Removing DPO or GRPO

generally results in lower Agreement and F1 than the full model, showing that debiasing-oriented preference optimization and consistency-oriented policy optimization both provide useful additional constraints. At the same time, some individual metrics exhibit non-monotonic behavior, such as the slightly higher Recall of W/O DPO on JudgeLM and the higher Precision of W/O GRPO on FairJudge-Benchmark-1K. Therefore, the ablation should not be interpreted as showing that one post-SFT stage uniformly dominates the others.

Overall, Table 2 supports the necessity of the staged design: SFT provides the core judging capability, while DPO and GRPO refine the judge toward more robust and consistent behavior. The full pipeline yields the most balanced performance across Agreement, Recall, and macro-F1, suggesting that FairJudge benefits from combining these supervision signals rather than relying on any single training objective.

4.3. Multimodal Evaluation: Judging Does Not Sacrifice Understanding

Table 3 reports results on multimodal benchmarks. FairJudge achieves competitive or superior accuracy compared with strong multimodal baselines, demonstrating that explicitly training a model as a judge does not degrade its multimodal understanding ability.

On tasks requiring both visual perception and high-level reasoning (e.g., InfographicsVQA and MathVista), FairJudge remains robust, indicating that judgment strategy learning can coexist with, and even complement, multimodal representation learning. This suggests that LLM-as-a-Judge models should be viewed as a distinct capability class rather than a weakened variant of generation-oriented models.

4.4. Consistency Analysis: Resolving Pointwise–Pairwise Conflicts

Table 4 directly evaluates consistency between pointwise and pairwise evaluation modes. FairJudge significantly outperforms all baselines on this metric, validating the effectiveness of explicitly enforcing cross-mode consistency during training. This result is particularly important in practice, as pointwise–pairwise inconsistency is a common yet under-explored failure mode of existing LLM-as-a-Judge systems. By modeling judging as a policy rather than an implicit function call, FairJudge substantially reduces contradictory decisions across evaluation settings.

4.5. Reward-Bench: Toward More Reliable Reward Signals

On Reward-Bench (Table 5), FairJudge achieves the best overall Agreement and macro-F1 among models of comparable scale. Compared with both general-purpose models

and prior judge-specific approaches, FairJudge provides more stable and consistent reward signals.

This property is especially desirable for downstream preference optimization and reinforcement learning, where unstable reward models often lead to training instability.

4.6. Inference Efficiency: Balancing Quality and Cost

Finally, Table 6 reports inference time comparisons between the Full and Fast modes. FairJudge achieves a consistent $12\times$ – $13\times$ speedup in Fast mode across different model scales, with only marginal performance degradation.

This efficiency makes FairJudge practical for large-scale automated evaluation and online assessment scenarios, where both reliability and throughput are critical.

Overall, FairJudge achieves a favorable balance among judgment quality, behavioral consistency, and inference efficiency, demonstrating its effectiveness as a general and deployable LLM-as-a-Judge framework.

5. Conclusion

We present FairJudge, a unified framework for LLM-as-a-Judge that mitigates evaluation bias and improves cross-mode consistency through structured data construction and staged training. Experimental results show that FairJudge achieves more stable and reliable automatic judgments across multiple benchmarks, while maintaining strong multimodal generalization and efficient inference.

Despite these gains, our evaluation does not exhaustively cover all possible rubrics, domains, or subtle bias types. FairJudge-Benchmark-1K is instance-disjoint from the training data but remains aligned with the same data construction pipeline, and larger-backbone training is left for future work. We therefore view FairJudge as a practical step toward more reliable automated evaluation, rather than a complete solution to all LLM-as-a-Judge reliability issues.

Acknowledgements

We thank the anonymous reviewers for their constructive feedback. This work was supported by the Zhejiang Provincial Natural Science Foundation of China (Grant No. LD24F030002).

Impact Statement

This work focuses on improving the reliability and consistency of LLM-as-a-Judge systems used for automatic evaluation of machine learning models. By reducing evaluation bias and inconsistency, FairJudge has the potential to improve the fairness, stability, and reproducibility of model

assessment in research and development settings.

The proposed method does not introduce new capabilities for content generation, nor does it directly interact with end users or sensitive personal data. As such, we do not anticipate significant negative societal impacts beyond those already associated with large language models. Potential misuse risks, such as over-reliance on automated evaluation without human oversight, can be mitigated by using FairJudge as a supporting tool rather than a sole decision maker.

Overall, we believe this work contributes positively to the responsible use of machine learning systems by promoting more reliable and transparent evaluation practices.

References

- Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., and Mané, D. Concrete problems in ai safety. *arXiv preprint arXiv:1606.06565*, 2016.
- Anghel, C., Anghel, A. A., Pecheanu, E., Cocu, A., Istrate, A., and Andrei, C. A. Diagnosing bias and instability in llm evaluation: A scalable pairwise meta-evaluator. *Information*, 16(8):652, 2025.
- Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang, K., Wang, P., Wang, S., Tang, J., et al. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., Das-Sarma, N., Drain, D., Fort, S., Ganguli, D., Henighan, T., et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.
- Bowman, S. R. Eight things to know about large language models. *Critical AI*, 2(2), 2024.
- Celikyilmaz, A., Clark, E., and Gao, J. Evaluation of text generation: A survey. *arXiv preprint arXiv:2006.14799*, 2020.
- Chen, D., Chen, R., Zhang, S., Wang, Y., Liu, Y., Zhou, H., Zhang, Q., Wan, Y., Zhou, P., and Sun, L. Mllm-as-a-judge: Assessing multimodal llm-as-a-judge with vision-language benchmark. In *Forty-first International Conference on Machine Learning*, 2024a.
- Chen, Z., Wu, J., Wang, W., Su, W., Chen, G., Xing, S., Zhong, M., Zhang, Q., Zhu, X., Lu, L., et al. Internvl: Scaling up vision foundation models and aligning for

- generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 24185–24198, 2024b.
- Chiang, W.-L., Zheng, L., Sheng, Y., Angelopoulos, A. N., Li, T., Li, D., Zhu, B., Zhang, H., Jordan, M., Gonzalez, J. E., et al. Chatbot arena: An open platform for evaluating llms by human preference. In *Forty-first International Conference on Machine Learning*, 2024.
- Christiano, P. F., Leike, J., Brown, T., Martic, M., Legg, S., and Amodei, D. Deep reinforcement learning from human preferences. *Advances in neural information processing systems*, 30, 2017.
- Christie, S. T., Moreau-Pernet, B., Tian, Y., and Whitmer, J. Flexeval: a customizable tool for chatbot performance evaluation and dialogue analysis. 2024.
- Cui, G., Yuan, L., Ding, N., Yao, G., He, B., Zhu, W., Ni, Y., Xie, G., Xie, R., Lin, Y., et al. Ultrafeedback: Boosting language models with scaled ai feedback. *arXiv preprint arXiv:2310.01377*, 2023.
- Dubois, Y., Galambosi, B., Liang, P., and Hashimoto, T. B. Length-controlled alpaca-eval: A simple way to debias automatic evaluators. *arXiv preprint arXiv:2404.04475*, 2024.
- Gao, L., Schulman, J., and Hilton, J. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, pp. 10835–10866. PMLR, 2023.
- Gehrmann, S., Adewumi, T., Aggarwal, K., Ammanamanchi, P. S., Aremu, A., Bosselut, A., Chandu, K. R., Clinciu, M., Das, D., Dhole, K., et al. The gem benchmark: Natural language generation, its evaluation and metrics. In *Proceedings of the 1st Workshop on Natural Language Generation, Evaluation, and Metrics (GEM 2021)*, pp. 96–120, 2021.
- Gilardi, F., Alizadeh, M., and Kubli, M. Chatgpt outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30): e2305016120, 2023.
- Gu, J., Jiang, X., Shi, Z., Tan, H., Zhai, X., Xu, C., Li, W., Shen, Y., Ma, S., Liu, H., et al. A survey on llm-as-a-judge. *The Innovation*, 2024.
- Hadfield-Menell, D., Milli, S., Abbeel, P., Russell, S. J., and Dragan, A. Inverse reward design. *Advances in neural information processing systems*, 30, 2017.
- Kim, S., Shin, J., Cho, Y., Jang, J., Longpre, S., Lee, H., Yun, S., Shin, S., Kim, S., Thorne, J., et al. Prometheus: Inducing fine-grained evaluation capability in language models. In *The Twelfth International Conference on Learning Representations*, 2023.
- Ko, J., Kim, S., Cho, S., and Yun, S.-Y. Flex-judge: Text-only reasoning unleashes zero-shot multimodal evaluators. *Advances in Neural Information Processing Systems*, 38:108720–108755, 2026.
- Kocmi, T. and Federmann, C. Large language models are state-of-the-art evaluators of translation quality. *arXiv preprint arXiv:2302.14520*, 2023.
- Lambert, N., Pyatkin, V., Morrison, J., Miranda, L. J. V., Lin, B. Y., Chandu, K., Dziri, N., Kumar, S., Zick, T., Choi, Y., et al. Rewardbench: Evaluating reward models for language modeling. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pp. 1755–1797, 2025.
- Li, D., Tan, Z., Zhao, C., Jiang, B., Huang, B., Ma, P., Alnaibari, A., Shu, K., and Liu, H. Who’s your judge? on the detectability of llm-generated judgments. *arXiv preprint arXiv:2509.25154*, 2025.
- Li, H., Dong, Q., Chen, J., Su, H., Zhou, Y., Ai, Q., Ye, Z., and Liu, Y. Llm-as-judges: a comprehensive survey on llm-based evaluation methods. *arXiv preprint arXiv:2412.05579*, 2024.
- Li, J., Sun, S., Yuan, W., Fan, R.-Z., Zhao, H., and Liu, P. Generative judge for evaluating alignment. *arXiv preprint arXiv:2310.05470*, 2023.
- Lin, C.-Y. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pp. 74–81, 2004.
- Liu, H., Li, C., Li, Y., and Lee, Y. J. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 26296–26306, 2024a.
- Liu, Y., Iter, D., Xu, Y., Wang, S., Xu, R., and Zhu, C. G-eval: Nlg evaluation using gpt-4 with better human alignment. *arXiv preprint arXiv:2303.16634*, 2023.
- Liu, Y., Moosavi, N. S., and Lin, C. Llm as narcissistic evaluators: When ego inflates evaluation scores. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 12688–12701, 2024b.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- Papineni, K., Roukos, S., Ward, T., and Zhu, W.-J. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pp. 311–318, 2002.

- Pombal, J., Yoon, D., Fernandes, P., Wu, I., Kim, S., Rei, R., Neubig, G., and Martins, A. F. M-prometheus: A suite of open multilingual llm judges. *arXiv preprint arXiv:2504.04953*, 2025.
- Rafailov, R., Sharma, A., Mitchell, E., Manning, C. D., Ermon, S., and Finn, C. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems*, 36: 53728–53741, 2023.
- Shi, L., Ma, C., Liang, W., Diao, X., Ma, W., and Vosoughi, S. Judging the judges: A systematic study of position bias in llm-as-a-judge. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pp. 292–314, 2025.
- Tripathi, T., Wadhwa, M., Durrett, G., and Niekum, S. Pairwise or pointwise? evaluating feedback protocols for bias in llm-based evaluation. *arXiv preprint arXiv:2504.14716*, 2025.
- Wang, P., Li, L., Chen, L., Cai, Z., Zhu, D., Lin, B., Cao, Y., Kong, L., Liu, Q., Liu, T., et al. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 9440–9450, 2024.
- Wang, Y., Yu, Z., Zeng, Z., Yang, L., Wang, C., Chen, H., Jiang, C., Xie, R., Wang, J., Xie, X., et al. Pandalm: An automatic evaluation benchmark for llm instruction tuning optimization. *arXiv preprint arXiv:2306.05087*, 2023.
- Wang, Y., Song, Y., Zhu, T., Zhang, X., Yu, Z., Chen, H., Song, C., Wang, Q., Wang, C., Wu, Z., et al. Trustjudge: Inconsistencies of llm-as-a-judge and how to alleviate them. *arXiv preprint arXiv:2509.21117*, 2025.
- Zhang, Y., Wang, C., Wu, L., Yu, W., Wang, Y., Bao, G., and Tang, J. Uda: Unsupervised debiasing alignment for pairwise llm-as-a-judge. *arXiv preprint arXiv:2508.09724*, 2025.
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36: 46595–46623, 2023.
- Zhou, H., Huang, H., Long, Y., Xu, B., Zhu, C., Cao, H., Yang, M., and Zhao, T. Mitigating the bias of large language model evaluation. In *China National Conference on Chinese Computational Linguistics*, pp. 451–462. Springer, 2024.
- Zhu, L., Wang, X., and Wang, X. Judgelm: Fine-tuned large language models are scalable judges. *arXiv preprint arXiv:2310.17631*, 2023.

A. Appendix

A.1. Direct Preference Optimization (DPO)

Direct Preference Optimization (DPO)

We construct a preference dataset $\mathcal{D}_{\text{pref}}$ consisting of tuples (q, y^+, y^-) , where q denotes the evaluation input (e.g., task, reference, answer pair, and evaluation context), and y^+ and y^- represent preferred and dispreferred judgment outputs, respectively. In our setting, preference pairs are constructed from bias-targeted evaluation instances. The preferred output y^+ corresponds to a rubric-consistent judgment that satisfies the desired invariance constraints under controlled non-semantic perturbations, while the dispreferred output y^- exhibits bias-related behaviors or violates these constraints. Teacher judgments are used as additional supervision signals to identify and verify high-quality preferences.

Following DPO, we define the probability that the policy π_ϕ prefers y^+ over y^- as a logistic model of the *relative log-likelihood ratio* against a fixed reference policy π_{ref} :

$$p_\phi(y^+ \succ y^- | q) = \sigma\left(\beta[\log \pi_\phi(y^+ | q) - \log \pi_\phi(y^- | q) - \log \pi_{\text{ref}}(y^+ | q) + \log \pi_{\text{ref}}(y^- | q)]\right), \quad (2)$$

where $\sigma(\cdot)$ is the sigmoid function and $\beta > 0$ is a temperature parameter.

The DPO objective minimizes the negative log-likelihood of the observed preferences:

$$\mathcal{L}_{\text{DPO}}(\phi) = \mathbb{E}_{(q, y^+, y^-) \sim \mathcal{D}_{\text{pref}}} [-\log p_\phi(y^+ \succ y^- | q)]. \quad (3)$$

Optimizing Eq. (3) increases the policy’s relative preference for rubric-consistent and bias-robust judgments while regularizing against excessive deviation from π_{ref} through the reference-normalized log-ratio in Eq. (2).

A.2. GRPO-Based Consistency Optimization

GRPO-Based Consistency Optimization

In the final training stage, we adopt Group-Relative Policy Optimization (GRPO) to further optimize the judging policy. For each evaluation input q , multiple candidate judgments are sampled and assigned scalar rewards. The rewards are normalized within each group to obtain group-relative advantages, which are then used to update the policy following the standard GRPO objective.

Rather than introducing a new policy-optimization algorithm, our contribution lies in the consistency-oriented reward design used during GRPO training. Specifically, aligned supervision signals from pointwise and pairwise evaluation settings are used to encourage judgments that remain compatible across evaluation modes. The detailed reward construction is described below.

A.3. Consistency-Oriented Reward Design

Consistency-Oriented Reward Design

To instantiate the reward signal used during GRPO training, we design a consistency-oriented hybrid reward function that adapts its scoring rule to different evaluation formats. Let c denote a model completion, $t \in \{\text{pp}, \text{sp_point}, \text{sp_pair}\}$ denote the task type, and g_1 and g_2 denote the corresponding ground-truth supervision signals. The reward is designed to encourage predictions that remain compatible with the same underlying preference relation across pointwise and pairwise evaluation settings. Specifically, `pp` evaluates whether a pairwise judgment agrees with the relative ordering implied by g_1 and g_2 , while `sp_point` and `sp_pair` evaluate agreement with their corresponding pointwise supervision signals. In this way, cross-mode consistency is encouraged through aligned ground-truth supervision rather than by directly matching the surface forms of outputs generated under different modes.

Pairwise Preference (pp). For pairwise comparison tasks, the model outputs a structured response containing specific sections (e.g., Rubric, Reasoning, Judgment). We employ a regex-based extractor $\mathcal{E}(\cdot)$ to locate the final decision token $\hat{y} \in \{\text{A_win}, \text{B_win}, \text{tie}\}$ within the ‘‘Judgment’’ section. The reward is binary: it assigns 2.0 if $\hat{y}$ is consistent with the numerical ground-truth scores g_1 and g_2 , and 0.0 otherwise.

Scalar Point Regression (sp_point). For numerical scoring tasks, we penalize the absolute deviation between the predicted integer value and the ground truth. The reward decays linearly with $|g_1 - \text{Int}(c)|$ and is clipped at zero.

Exact Matching (sp_pair). For short-answer or label generation, we apply strict exact matching.

The unified reward function $R(c, t, g_1, g_2)$ is defined as:

$$R(c, t, g_1, g_2) = \begin{cases} 2.0 \cdot \mathbb{I}[\text{Consistent}(\mathcal{E}(c), g_1, g_2)], & \text{if } t = \text{pp}, \\ \max(0, 2 - |g_1 - \text{Int}(c)|), & \text{if } t = \text{sp_point}, \\ 2.0 \cdot \mathbb{I}[c = g_1], & \text{if } t = \text{sp_pair}, \\ 0.0, & \text{otherwise.} \end{cases} \quad (4)$$

where $\mathbb{I}[\cdot]$ is the indicator function. The consistency predicate is defined as:

$$\text{Consistent}(\hat{y}, g_1, g_2) \iff ((\hat{y} = \text{A_win} \wedge g_1 > g_2) \vee (\hat{y} = \text{B_win} \wedge g_2 > g_1) \vee (\hat{y} = \text{tie} \wedge g_1 = g_2)). \quad (5)$$

Algorithm 1: Consistent Hybrid Reward Calculation

Require: Completion c , Task Type t , Ground Truths g_1, g_2

Ensure: Reward $r \in [0, 2]$

```

 $r \leftarrow 0.0$ 
if parsing succeeds then
    if  $t = \text{pp}$  then
        Extract judgment section from  $c$  using regex
        Parse label  $\hat{y} \in \{\text{A\_win}, \text{B\_win}, \text{tie}\}$ 
        if  $g_1 > g_2$  and  $\hat{y} = \text{A\_win}$  then  $r \leftarrow 2.0$ 
        else if  $g_2 > g_1$  and  $\hat{y} = \text{B\_win}$  then  $r \leftarrow 2.0$ 
        else if  $g_1 = g_2$  and  $\hat{y} = \text{tie}$  then  $r \leftarrow 2.0$ 
        end if
    else if  $t = \text{sp\_point}$  then
         $v \leftarrow \text{Int}(c)$ 
         $r \leftarrow \max(2.0 - |g_1 - v|, 0.0)$ 
    else if  $t = \text{sp\_pair}$  then
        if  $c = g_1$  then  $r \leftarrow 2.0$ 
        end if
    end if
else
     $r \leftarrow 0.0$ 
end if
return  $r$ 
    
```

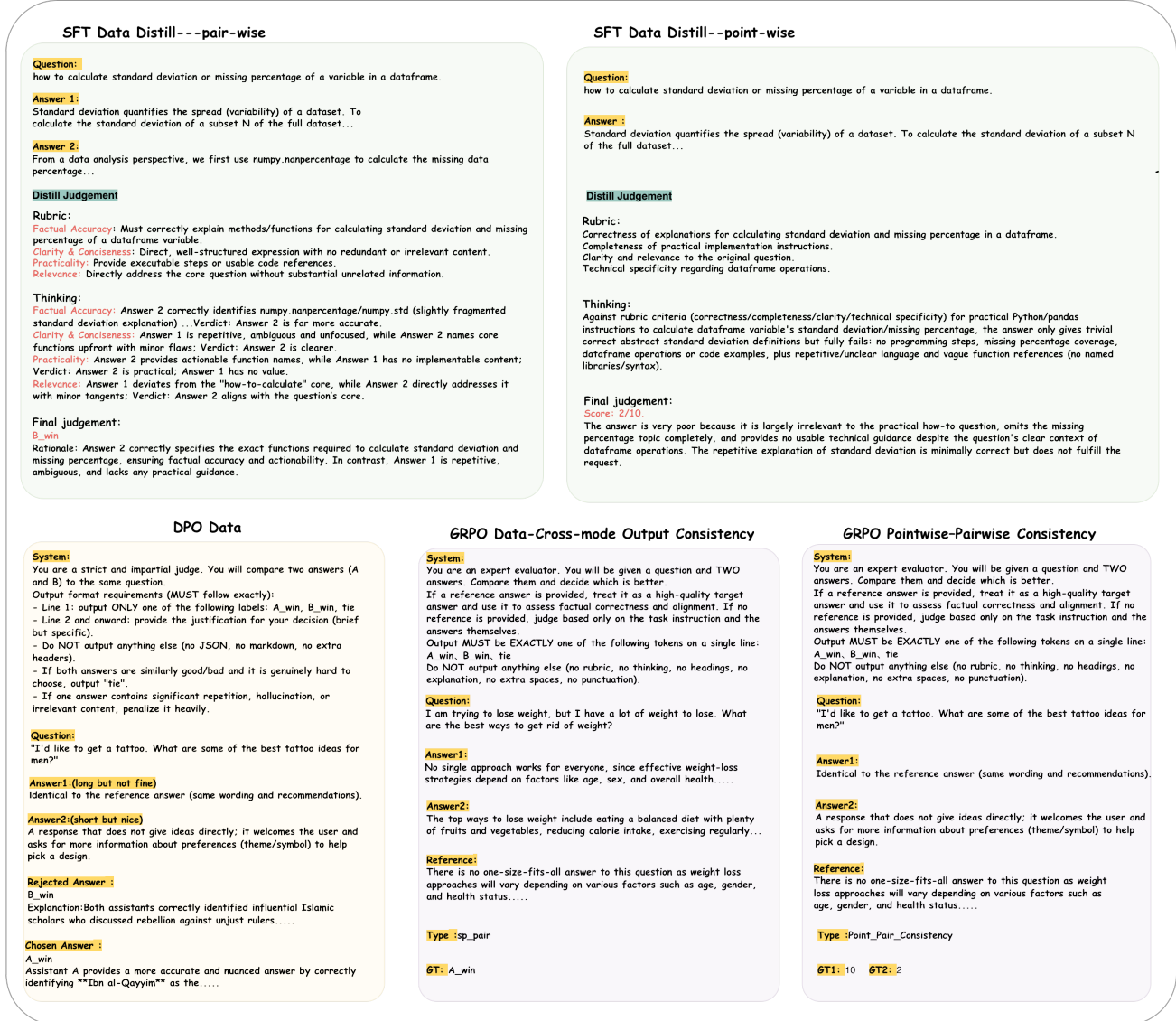


Figure 5. Representative data formats used in different training stages of FairJudge, including distilled judgment examples for supervised fine-tuning (SFT) (top: pair-wise and point-wise), as well as training data for preference alignment and consistency optimization via DPO and GRPO (bottom).

DeepseekV3 REAL Failure Case Show

Question:

Rewrite the following paragraph using fewer words and no loss of meaning.\nManufacturers all around the globe are fighting a battle to reduce their waste and increase the sustainability of their production.

Answer1 :

Manufacturers are working hard to reduce waste and increase the sustainability of their production.

Answer2 :

Manufacturers are constantly striving to reduce waste and increase sustainability in their production processes.\n\nInstruction:

DeepSeek Judgement

DeepSeek Answer2 Pointwise judgement:

Score: 8

The answer successfully reduces the word count and retains the essential meaning. The loss of \"all around the globe\" slightly reduces specificity but does not constitute a loss of core meaning, and the substitution for \"fighting a battle\" is acceptable. It is a good, effective rewrite.

DeepSeek Answer1 Pointwise judgement:

Score: 6

Justification: The answer is significantly shorter and captures the main actions, but it loses the important detail about the effort being global, resulting in a partial loss of the original meaning. This places it in the average category.",

DeepSeek Pairwise judgement:

tie

Both answers successfully reduce word count by similar margins while preserving meaning and clarity. The minor differences in phrasing do not meaningfully advantage one over the other."

Figure 6. A real cross-mode inconsistency case of DeepSeek-V3. The example is drawn from real evaluation data. Although DeepSeek-V3 assigns different quality scores to the two answers in pointwise evaluation, it predicts a tie in pairwise evaluation, revealing inconsistent judgments across evaluation modes.

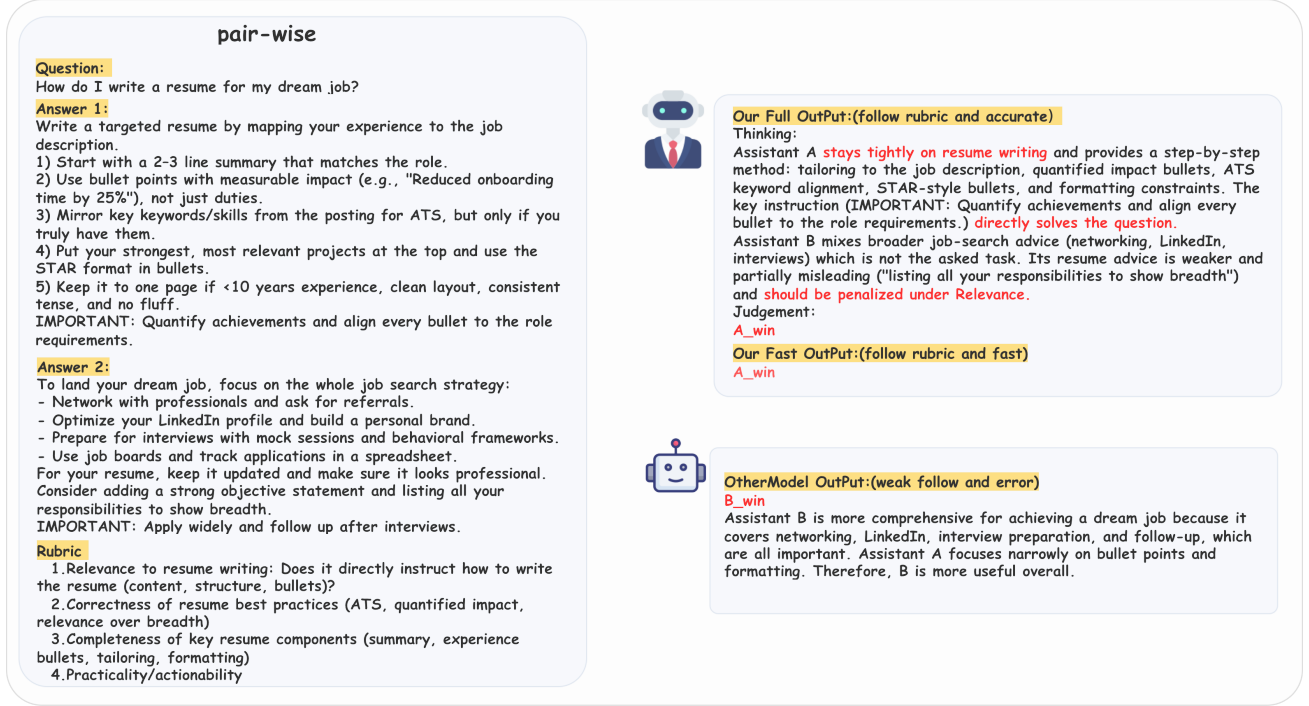


Figure 7. Comparison between FairJudge and a baseline model on real evaluation data. Under the same input and unified evaluation rubric, **FairJudge** strictly follows task instructions and judging criteria, correctly selecting *Answer A* that is highly aligned with the query. The figure also illustrates our two output modes—*full reasoning output* and *fast decision output*—which yield consistent judgments while balancing interpretability and efficiency. In contrast, the baseline model overemphasizes surface-level coverage and generic advice, incorrectly favoring *Answer B* despite its inclusion of task-irrelevant content, leading to a suboptimal decision.

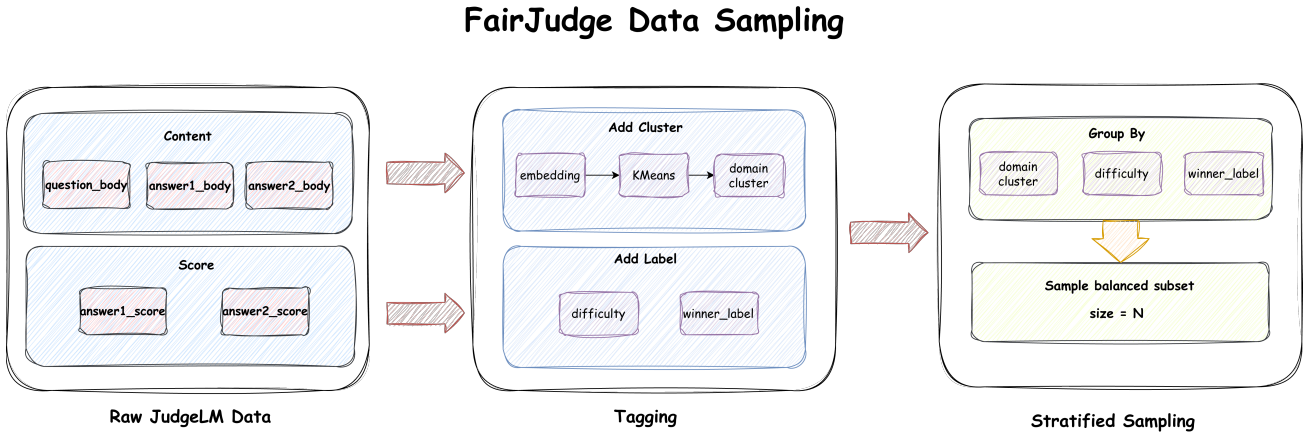


Figure 8. The figure illustrates the FairJudge data sampling pipeline. Starting from raw JudgeLM preference data, we first extract the core content fields (question, answer₁, answer₂) together with their associated scores (answer₁_score, answer₂_score). In the tagging stage, each instance is assigned to a domain cluster by computing text embeddings and applying KMeans clustering, and supervision signals—including a difficulty label and a winner label—are derived from score-based comparisons. Finally, we perform stratified sampling over the Cartesian product of $domain_cluster \times difficulty \times winner_label$ to construct a balanced subset of size N , which improves representativeness and diversity while mitigating label-distribution bias and filtering low-quality samples for training.

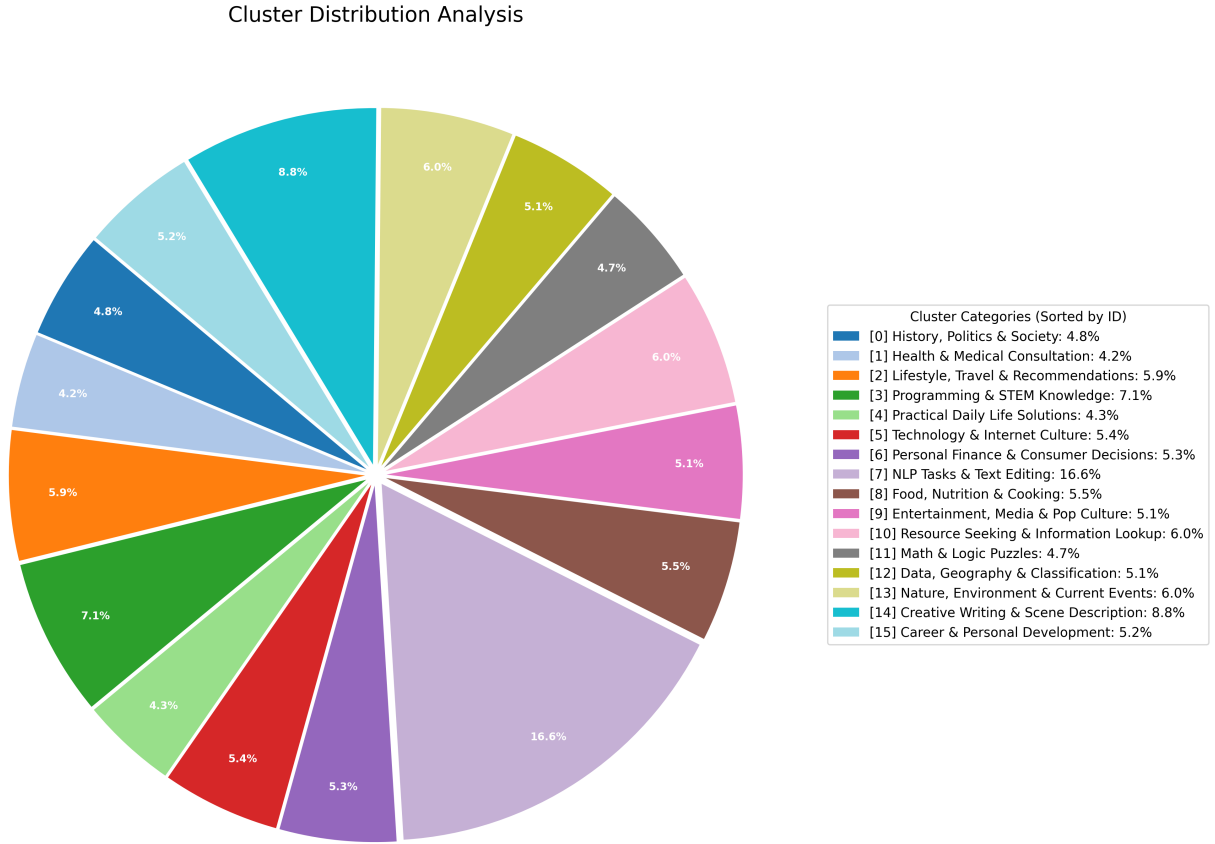


Figure 9. Cluster distribution of the sampled FairJudge data. The figure shows the semantic domain distribution obtained during the data sampling process. The sampled data covers 16 diverse clusters, including NLP tasks and text editing, creative writing and scene description, programming and STEM knowledge, resource seeking and information lookup, health and medical consultation, and other common instruction-following domains. The largest cluster is NLP Tasks & Text Editing (16.6%), followed by Creative Writing & Scene Description (8.8%) and Programming & STEM Knowledge (7.1%), while most other clusters remain within a moderate range of approximately 4–6%. This distribution indicates that the sampling process preserves broad domain coverage and avoids excessive concentration in a single topic category.

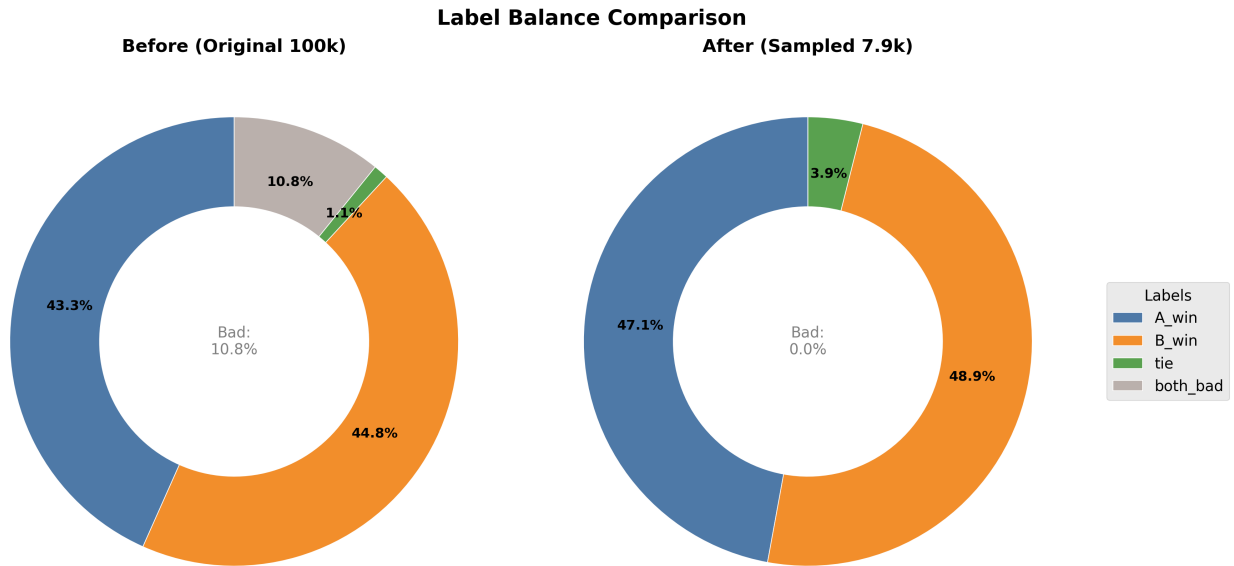


Figure 10. Label distribution comparison before (Original 100k) and after sampling (Sampled 7.9k). The figure shows that the sampling process largely preserves the balance between A_win and B_win labels while substantially improving data quality by removing low-quality samples. In the original dataset, A_win and B_win account for 43.3% and 44.8%, respectively, with tie at 1.1% and both_bad (invalid or low-quality comparisons) at 10.8%. After sampling, A_win and B_win slightly increase to 47.1% and 48.9%, tie rises to 3.9%, and both_bad is completely eliminated (0.0%). This indicates that the sampling strategy effectively removes a substantial portion of low-quality comparisons without introducing noticeable class bias, thereby improving the purity of supervision signals. Meanwhile, the increased proportion of tie samples suggests a higher presence of fine-grained or borderline preference cases, which can help the model better learn subtle preference distinctions.

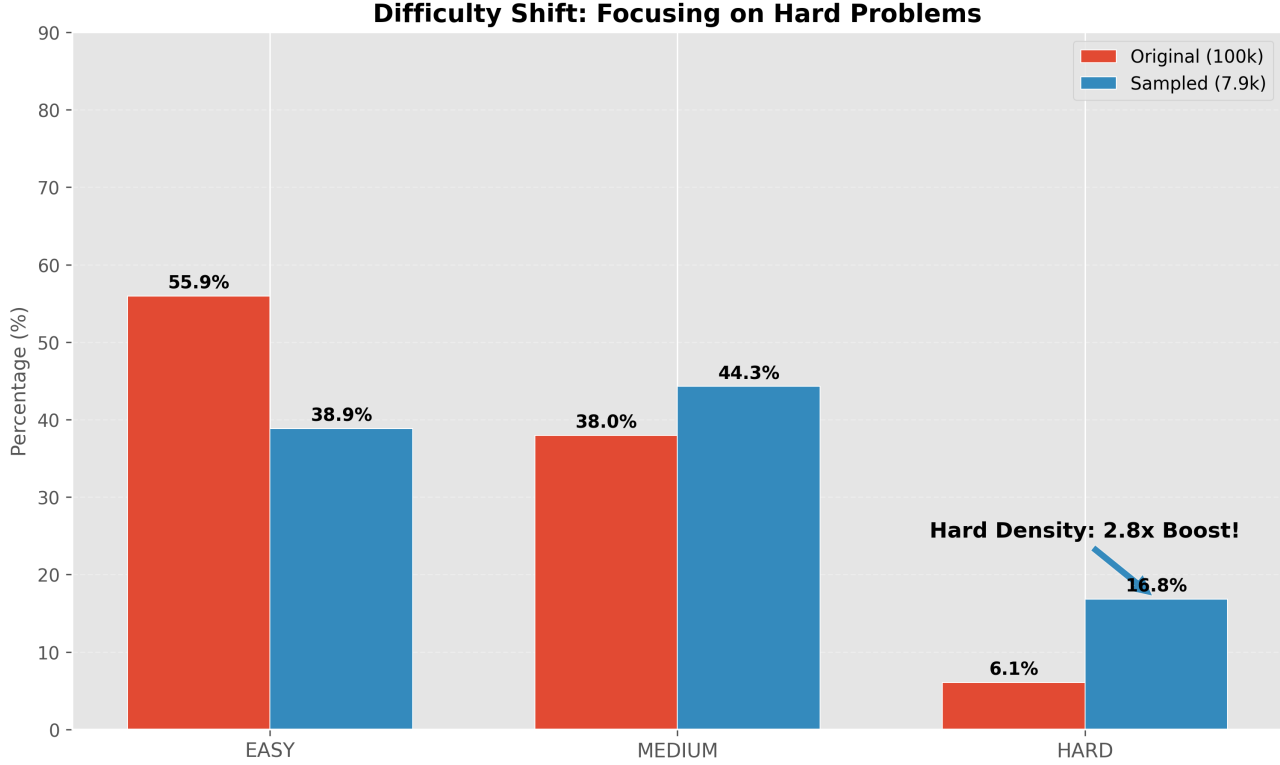


Figure 11. Difficulty distribution shift induced by data sampling. The figure illustrates the change in difficulty composition from the original 100k dataset to the sampled 7.9k subset. In the original data, EASY / MEDIUM / HARD samples account for 55.9%, 38.0%, and 6.1%, respectively, whereas the sampled set exhibits a distribution of 38.9% / 44.3% / 16.8%. Compared to the original distribution, the proportion of EASY samples decreases by 17.0 percentage points, while MEDIUM and HARD increase by 6.3 and 10.7 points, respectively. Notably, the relative density of HARD samples is boosted by approximately $2.8\times$ ($16.8\%/6.1\% \approx 2.8$), indicating that the sampling strategy effectively shifts the training focus toward more challenging instances, thereby increasing the information density for learning under complex scenarios.